

**Foundation (Jalapeno)**

- Q1.** What is the point-slope form of a line with slope  $m = 2$  and passing through the point  $(1, 3)$ ?
- (A)  $y - 3 = 2(x - 1)$   
 (B)  $y - 1 = 2(x - 3)$   
 (C)  $y - 2 = 3(x - 1)$   
 (D)  $y - 1 = 3(x - 2)$   
 (E)  $y - 3 = 1(x - 2)$
- Q2.** A sprinter runs 100 meters in 10 seconds. If she maintains a constant speed, what is her distance  $d$  after  $t$  seconds in slope-intercept form?
- (A)  $d = 10t + 0$   
 (B)  $d = 10t - 5$   
 (C)  $d = 5t + 10$   
 (D)  $d = 5t - 10$   
 (E)  $d = 0t + 100$
- Q3.** The equation of a line in standard form is  $2x + 3y = 12$ . What is the  $y$ -intercept?
- (A)  $y = 4$   
 (B)  $y = -4$   
 (C)  $y = 2$   
 (D)  $y = -2$   
 (E)  $y = 6$
- Q4.** A basketball player scores an average of 20 points per game. If she has played  $g$  games, what is her total score  $s$  in point-slope form?
- (A)  $s - 0 = 20(g - 0)$   
 (B)  $s - 10 = 20(g - 1)$   
 (C)  $s - 20 = 10(g - 2)$   
 (D)  $s - 0 = 10(g - 0)$   
 (E)  $s - 20 = 20(g - 1)$
- Q5.** The slope-intercept form of a line is  $y = -2x + 5$ . What is the equation of this line in standard form?
- (A)  $2x + y = 5$   
 (B)  $2x + y = -5$   
 (C)  $2x - y = 5$   
 (D)  $2x - y = -5$   
 (E)  $x + 2y = 5$
- Q6.** A cyclist rides 25 miles in 2 hours. If she maintains a constant speed, what is her distance  $d$  after  $t$  hours in slope-intercept form?
- (A)  $d = 12.5t + 0$   
 (B)  $d = 12.5t - 5$   
 (C)  $d = 10t + 5$   
 (D)  $d = 10t - 5$   
 (E)  $d = 5t + 10$
- Q7.** The equation of a line in point-slope form is  $y - 2 = 3(x - 1)$ . What is the  $x$ -intercept?
- (A)  $x = -1$   
 (B)  $x = 1$   
 (C)  $x = -2$   
 (D)  $x = 2$   
 (E)  $x = 0$
- Q8.** A tennis player wins an average of 30 points per set. If she has played  $s$  sets, what is her total score  $t$  in slope-intercept form?
- (A)  $t = 30s + 0$   
 (B)  $t = 30s - 10$   
 (C)  $t = 10s + 30$   
 (D)  $t = 10s - 30$   
 (E)  $t = 0s + 30$

**Open Ended Questions (FRQ)**

- Q9.** A swimmer swims 200 meters in 8 minutes. If she maintains a constant speed, find the equation of the line that represents her distance  $d$  after  $t$  minutes in standard form. Then, graph the equation and find the  $d$ -intercept.
- Q10.** The equation of a line in slope-intercept form is  $y = -4x + 20$ . Convert this equation to point-slope form and then graph it. Find the  $y$ -intercept and the  $x$ -intercept.

**Chef's Tips**

- Tip 1: Emphasize the importance of labeling axes and title when graphing.
- Tip 2: Encourage students to check their work by plugging in values.
- Tip 3: Remind students to show all work and explain their reasoning for free response questions.